

Method Overloading :

Question 1 :

Define a class to overload the method display() as follows:

`void display(int r, int c)` - To print a pattern.

If r is 3, and c is 4, the following pattern is printed:

```
1 1 1 1
2 2 2 2
3 3 3 3
```

`void display(int n)` - To find and print the product of n terms for a series.

If n is 5, the product will be $1 \times 3 \times 5 \times 7 \times 9$

`void display(double h, double r)` - To find and print the total surface area of a cylinder.

$$\text{Surface area of a cylinder} = 2\pi r(h + r)$$

Question 2 :

Define a class to overload the method print() as follows:

`boolean print(int n)` - To check if a number is a Dudeney number. A number is said to be a Dudeney number if the cube root of the number is equal to the sum of the digits of the number. The method should return true if the number is a Dudeney number, and false otherwise.

$$\sqrt[3]{512} = 8 = 5 + 1 + 2 \rightarrow 512 \text{ is a Dudeney number}$$

`void print(double p, int t, double r)` - To find and print the simple interest from the principal amount, time, and rate of interest. Use the following formula for simple interest.

$$S.I. = \frac{p \times t \times r}{100}$$

`double print(int m, int n, boolean b)` - To find the sum of m and n if b is true, and to find the quotient of m and n if b is false. Print the appropriate answer.

Question 3 :

Define a class to overload a method perform() as follows

`boolean perform(int n)` - To check if a number is a Duck number or not. A number is said to be a Duck number if it has zeroes present in it, but doesn't have a zero at the beginning of the number. The method should return true if the number is a Duck number, and false otherwise.

`double perform(double a, int c)` - To find and return the sum of a number of c terms in a series with a difference of 0.2 starting from a.

Sum = 3.1 + 3.3 + 3.5 + 3.7 + 3.9 + 4.1

`char perform(int a, int b)` - To add two numbers and find the character equivalent of the resultant sum. Return the character formed.

If $a = 30$, and $b = 35 \rightarrow a + b = 65 = 'A'$

Question 4 :

Define a class to overload a method num() as follows:

`void num(int r, int c):` To print a pattern with r number of rows and c number of columns.

When r = 4, and c = 5

```
1 0 1 0 1
0 1 0 1 0
1 0 1 0 1
0 1 0 1 0
```

`int num(int x) -` To return the sum of the following series:

$$\frac{x}{1} + \frac{x}{8} + \frac{x}{27} + \frac{x}{64} + \frac{x}{125} + \frac{x}{216} + \frac{x}{343}$$

`double num(double b, double h) -` To calculate the area of a triangle with base b and height h.

$$\text{Area of Triangle} = \frac{1}{2} \times b \times h$$